

## 4. Results and Discussion

### 4.1 Overview of Experimental Setup

All six models considered in this paper ResNet50, MobileNetV2, DenseNet121, EfficientNetB0, a Custom CNN, and the proposed CBAM-enhanced attention CNN were trained and tested on the augmented Alzheimer MRI dataset using a strict 5-fold cross-validation strategy. The entire pool of 33,984 training images (four classes: Non Demented, Very Mild Demented, Mild Demented and Moderate Demented) was split so that, in each fold, 90% of the non-test pool was used as training, and 10% as validation, and the other fold was an entirely unknown hold-out test set. Each of the models was trained to a maximum of 50 epochs with the Adam optimizer with a learning rate of  $1 \times 10^{-4}$  and early stopping (patience = 10) based on validation accuracy to avoid overfitting. All of it was done with dual T4 parallelism with the Mirrored Strategy in TensorFlow, and a global random seed of 42 was used to achieve complete reproducibility. Reported metrics of evaluation here accuracy, weighted F1 score, precision and recall were calculated on the hold out test fold of each split and the average of all five folds.

### 4.2 Overall Performance Comparison

Table 1 is a summary of all the four metrics mean  $\pm$  standard deviation on five folds of each model according to the composite score (unweighted average of the four metrics). The proposed CBAM CNN has the best performance in all reported metrics reaching a mean test accuracy of 93.71 percent and weighted F1 score of 96.70 percent. These scores are a significant boost over the next-best pretrained model, ResNet50 (92.87% accuracy), and a significant improvement over the baseline Custom CNN in Task 3 (72.59% accuracy) even though they both have the same backbone architecture with only the addition of CBAM attention blocks.

Table 1. 5-Fold Cross-Validation Performance Comparison of All Six Models (Mean  $\pm$  Std Dev). Best results in bold and highlighted.

| Model                  | Accuracy (%)                       | F1 Score (%)                       | Precision (%)                      | Recall (%)                         | Composite (%) |
|------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|---------------|
| <b>CBAM CNN (Ours)</b> | <b>93.71 <math>\pm</math> 0.86</b> | <b>93.70 <math>\pm</math> 0.85</b> | <b>93.79 <math>\pm</math> 0.88</b> | <b>93.71 <math>\pm</math> 0.86</b> | <b>93.73</b>  |
| Custom CNN             | 72.59 $\pm$ 5.27                   | 71.37 $\pm$ 5.25                   | 76.68 $\pm$ 5.58                   | 72.59 $\pm$ 5.27                   | 73.31         |
| EfficientNetB0         | 88.00 $\pm$ 0.27                   | 87.97 $\pm$ 0.28                   | 87.98 $\pm$ 0.30                   | 88.00 $\pm$ 0.27                   | 87.99         |
| DenseNet121            | 87.48 $\pm$ 0.42                   | 87.43 $\pm$ 0.42                   | 87.48 $\pm$ 0.40                   | 87.48 $\pm$ 0.42                   | 87.47         |
| MobileNetV2            | 91.36 $\pm$ 0.07                   | 91.34 $\pm$ 0.07                   | 91.41 $\pm$ 0.04                   | 91.36 $\pm$ 0.07                   | 91.37         |
| ResNet50               | 92.87 $\pm$ 0.06                   | 92.87 $\pm$ 0.06                   | 92.92 $\pm$ 0.07                   | 92.87 $\pm$ 0.06                   | 92.88         |

Among the pretrained transfer learning baselines, ResNet50 emerged as the strongest at 92.88% accuracy, followed closely by MobileNetV2 at 91.37%. EfficientNetB0 (87.99%) and

DenseNet121 (87.47%) performed comparatively lower, which is consistent with prior literature — ResNet50's deep residual architecture, though powerful on natural image benchmarks, tends to overfit or underperform on narrow-domain medical datasets when feature extraction layers are fully frozen, as was the case here. MobileNetV2's lower parameter count, designed for mobile efficiency, limits its representational capacity for subtle inter-class differences in MRI texture.

Notably, the baseline Custom CNN — trained from scratch with just four convolutional blocks — managed to achieved accuracy of 73.31%.

### 4.3 Ablation Study: Effect of CBAM Attention

Since the CBAM CNN and the Custom CNN share an identical backbone — four convolutional blocks with filters of size 32, 64, 128, and 256, followed by global average pooling, a 256-unit dense layer, and 0.5 dropout — the gap in performance between them constitutes a direct, controlled **ablation study** on the contribution of the Convolutional Block Attention Module. Table 2 reports the per-fold accuracy and F1 score for both models, making this comparison transparent.

Table 2. Per-Fold Ablation (approximately): Custom CNN (without CBAM) vs. CBAM CNN (with CBAM).  $\Delta$  denotes the per-fold absolute gain in accuracy.

| Fold   | Custom CNN Acc. (%) | CBAM CNN Acc. (%) | $\Delta$ Gain (pp) | Custom CNN F1 (%) | CBAM CNN F1 (%) |
|--------|---------------------|-------------------|--------------------|-------------------|-----------------|
| Fold 1 | 79.5%               | 93.3%             | +13.8              | 76                | 93.3            |
| Fold 2 | 69.5%               | 93.8%             | +24.3              | 66.8              | 93.8            |
| Fold 3 | 75%                 | 94%               | +19                | 76                | 94              |
| Fold 4 | 66.05%              | 92.55%            | +26.5              | 64                | 93              |
| Fold 5 | 76.5%               | 95%               | +18.5              | 75                | 95              |
| Mean   | 73.31               | 93.73             | +20.42             | 71.56             | 93.82           |

The CBAM CNN consistently outperformed the baseline CNN model across all five folds, with per-fold accuracy gains. Importantly, the standard deviation of the CBAM CNN is noticeably lower than that of the Custom CNN, indicating that attention-guided feature selection not only lifts performance but also stabilizes it across different data partitions — a critical property for clinical deployment scenarios where prediction consistency is as important as peak accuracy.

This result aligns with the theoretical motivation of CBAM and spatial attention (which spatial regions matter?), the model learns to suppress uninformative background signals in the skull-stripped MRI scans and focus on clinically relevant regions such as the hippocampus, entorhinal cortex, and ventricular spaces, all of which are known biomarkers of neurodegeneration. The MRI images in this dataset contain axial brain slices with a pure black background; channel attention is particularly effective in this setting because it can down-weight feature maps that activate primarily on background noise while amplifying maps sensitive to cortical atrophy and ventricular enlargement — the core discriminative signals between the four dementia stages.

## 4.5 Cross-Validation Stability and Generalization

The main concept behind the validity of the reported results is the usage of 5-fold stratified cross-validation instead of one train-test split. Cycling hold-out partition on all five splits means that each image of the dataset gets used in both training and evaluation only once, which gives a low-variance estimate of generalization performance that is especially crucial in the present dataset due to the synthetic augmentation. With only one random seed, splitting may accidentally test images that are close to training images (a common risk with offline augmentation pipelines), and inflate accuracy reports. This is avoided by the stratified k-fold design, which guarantees that each class is represented proportionately in the test partition of each fold. The standard deviations of the CBAM CNN are always low (accuracy  $\pm 0.86\%$ , F1  $\pm 0.85\%$ , precision  $\pm 0.88$ , recall  $\pm 0.86$ ) indicating that the performance of the model is consistent and not due to a favorable data split.

## 4.6 Discussion

Taken together, the results establish three key contributions of this work. First, they demonstrate that established pretrained transfer learning models (ResNet50, MobileNetV2, DenseNet121, EfficientNetB0) outperform the Custom CNN model but CBAM outperformed all the models on this Alzheimer's MRI classification task. This challenges the assumption that ImageNet pre-training is universally beneficial for medical imaging, particularly when the source and target domains are as different as natural photographs and grayscale axial brain scans.

Second, the ablation study provides rigorous, controlled evidence that CBAM attention is the direct cause of the Custom CNN's performance improvement — not differences in depth, capacity, optimizer, regularization, or data splits. This makes the contribution scientifically falsifiable and reproducible, which are prerequisites for publication in high-impact venues.

Third, the per-class analysis reveals that the hardest classification boundary remains between Mild and Moderate Demented stages, which mirrors the clinical challenge of distinguishing these stages from MRI scans alone. This points towards future directions: incorporating temporal imaging data, multi-modal fusion with CSF biomarkers, or class-weighted loss functions that specifically penalize Mild/Moderate confusion may push performance further. Self-supervised pre-training on brain MRI corpora (rather than ImageNet) could also be explored as an alternative to the augmentation-heavy approach used here.

One limitation worth acknowledging is that the dataset used here is augmented, meaning the effective diversity of the training distribution is constrained by the augmentation strategy applied during curation. While the 5-fold protocol mitigates data leakage risk, future work should validate the proposed CBAM CNN on a fully independent, prospective clinical dataset to confirm that the observed gains hold in practice. Nonetheless, within the scope of this study, the proposed architecture demonstrates clear, consistent, and reproducible superiority over all competing baselines.